

Design Compliance

This discussion will start with the design compliance of a larger system and then dive deeper into its lower level subsystems. Design compliance will be verified with respect to actual specifications set out at the start of the project, accessibility of design choice, cost of design choice and overall final functionality.

Mechanical

First and foremost, the scope of the quadruped had to be reduced from a 12 degree of freedom (DoF) quadruped to a 8 DoF quadruped because of weight / motor unreliability issues that will be discussed in the implementation section.

Body

The final body / shells were FDM printed using PLA. Although this choice was inline with structural analysis and accessibility requirements; the heat from overloaded motors deforms the structure and dimensional inaccuracies throws off belt tensioning / centering motor shaft in encoder.

The shell was supposed to be printed in ABS-PA but was ignored due to time constraints, prototyping and testing with PLA / PETG was quick as all team members could print with these filaments.

Actuator

With our choice of motor and transmission, the final actuator weighed in at 220-240 grams depending on the degree of freedom it controlled; which overshoots the targeted weight by almost 55% and leads the overall weight of the dog into the 3.3 kg range with offloaded electronics.

The motor was supposed to operate up to a maximum of 20A and this 33% increase in peak current should have partially offset the increase in weight with the final design cost being a less dynamic quadruped.

Since the motors regularly go for 25\$ on Aliexpress; this was thought of as a nice cost vs performance sacrifice at the time. The final actuator torque was inline with our initial design spec; if the motors did not tend to overheat at current values below its peak.

Transmission

The actuator featured a two stage 9:1 timing belt reduction featuring S3M timing belts which are a much more accessible alternative to the Contitechs in the original design and also cost significantly less.

Great time was spent tuning the tensioning of these belts to our satisfaction. The belts were sized to our power requirement and worked well. The belts overall are hard for assembly and not accessible to design around / modify.

The smaller resin printed pulley at the beginning of the second reduction stage would crack and powderise and hence needed to be replaced with a sintered Aluminum version, and this hurt the accessibility of the transmission since this part needs access to a SLS printer to be manufactured.

Motors

Despite promising specifications, the motors turned out to be very unreliable with respect to power output; details outlined in the implementation section.

Overall the mechanical system did its job but struggled with dynamic movements, overloaded belts, manufacturing tolerances that would mess with either tensioning or encoder misalignments and general robustness. *This leads into the realisation that making a product truly accessible involves significant time and monetary investment initially;* these efforts are not particularly at the forefront once the product reaches accessible status.

Mechanical Design Compliance Assessment

Objective	Status		
	Achieved	Partially Achieved	Not Achieved
9:1 Transmission System	x		
Robust Chassis		x	
2.8 kg Quadraped			x
	Final weight ~3.4 kgs with offloaded electronics		
12 DoF Quadraped			x
	Reduced to 8 DoF		

2.7 N-m Stall Torque at Actuator Output	~2.7-3 N-m range, would have preferred a higher initial specification		
Backdrivable Actuator	x		
Robust Transmission		x	
	Prefer a higher safety factor with respect to maximum power delivery of belts; source wider belts		
Modular Design	x		
Accessible Manufacturing		x	

Table 1. Mechanical Compliance Assessment

Mechanical Design

Pulleys

Pulleys printed with ABS-like resin were tested in the first iteration of the actuators. These pulleys brought with them many points of uncertainties which popped up during their testing:

- ABS-like resin is still far from an injection molded / sintered ABS part, the ABS-like resin is still very brittle
- The maximum bending stress at the root of the pulley was ~80% yield stress of the material
- The epoxy used to these pulleys to the shaft was not applied perfectly every time; lack of quality control during assembly

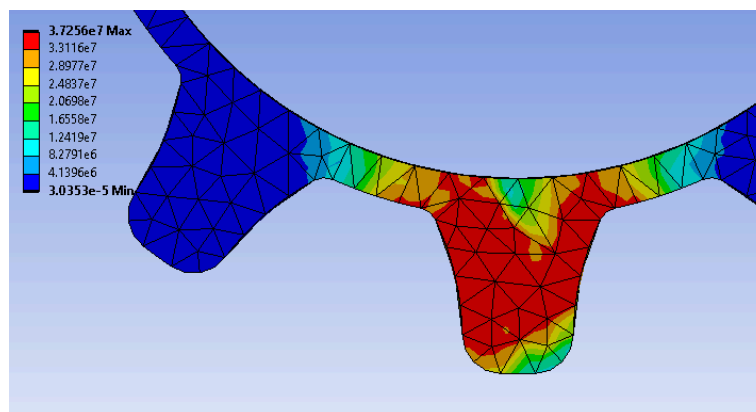


Figure 6. Pulley Tooth Stress Contour

Although the small 10-tooth pulleys sufficed for no-load testing; they were too unreliable and prone to failure to be included in the final design. Whereas the larger 30-tooth pulleys did not fail and were used even in the final assembly.

Also the small wall thickness of the 10-tooth pulleys meant ideally they were manufactured as a feature on the center shaft. Hence, an engineering decision was made to get all the pulleys and motor shafts sintered from JLC for the final design. In aiming for accessibility, these designs are hosted on JLC's model database and can be ordered on demand.

The SLS shaft diameters were undersized by 0.1mm which introduced concentricity issues with the motors rotor and stator; flat 3D printed sheets were used as shims in the housing to remove play and account for clearance.

Belts

Standard S3M belts of length 150mm and 210mm were sourced; which differ slightly from those used by the original Solo12 which meant re-tuning center distances to get appropriate tensioning. See [Appendix 1 : Belt Sizing](#) for belt sizing calculations.

Unlike the original design which uses sintered actuator shells, multiple iterations with shaft locations at intervals of 0.1mm in both x and y directions with respect to the theoretical location in the CAD were tested before settling on tensioner locations.



Figure 7. Tensioner Location Testing Jig

The tensioner locations needed to be re-tuned again once the sintered shafts arrived from JLC since they were undersized diametrically.

Actuator Shells

To account for the addition of the AMT103 encoder, S3M timing belts, Tarot 5008 motors, belt tensioners, M6 motor shaft and M4 center shaft the actuator shells had to go through many iterations, mainly due to the following reasons:

- New off-the-shelf components in the design

- Tolerance studies with respect to our requirement and the tolerance of our manufacturing method
- Design changes with respect to important locating features inside the shell
- Design for assembly improvements after an actuator was assembled



Figure 8. Box of Shell Prototypes

Body

The quadruped body underwent the following design changes:

- New Top / Bottom Center plates with Load Analysis
 - Center plates updated to accommodate bulky adduction Tarot motors
 - Addition of IMU mount holes
 - Addition of Realsense mount
- Updated side plates for driver cages (Driver cages not included in final design)
- Updated hip supports
- Updated front / back hip supports to attach to jig

Everytime an actuator shell underwent thickness changes; clearances needed to be verified with respect to the overall assembly. Also everytime an actuator was lengthened; center plates. side supports, hip supports needed to be changed as well and this added to the design time initially.

Manufacturing & Dissemination

Following the project's goal, we targeted an aggressive BoM price of 3500 USD. In order to achieve it, we prioritized off-the-shelf parts (drivers, motors, etc.) and minimized custom manufactured parts (actuator gears, shafts, etc.). The following table details exact costs of manufacturing one Newton at the time of writing and we expect that if it were productized at a later date, costs could easily come down on various components through economies of scale and hopefully less turbulent international trade conditions.

Item	Cost (USD)
12x Tarot 5008/340KV Brushless Motor	432.00
12x AMT103-V Encoder	361.00
12x ODrive S1 ¹	1788.00
Intel Realsense Depth Camera D455i	195.00
SYD Dynamics TM171	58.00
Jetson Orin Nano	249.00
LM2596 Voltage Converter	1.50
24x S3M Timing Belts	237.00
SLS Shafts & Gears	327.00
2x 6S Lipo Battery	109.00
Ethernet Cables	58.00
Various Wires	45.00
Nuts, Bolts and Fasteners	45.00
3D Printing Material	95.00
Total	3,999.50

Table 5. Production Bill of Materials

¹ This item could come down an additional 10% if an educational institution purchases them, since, at the time of writing, ODrive offers an educational discount.

Final vs Proposed Design Reflection

Mechanical

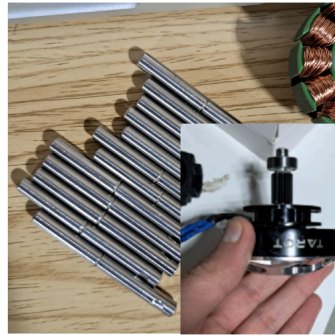
Motors were initially sized for a ~3 kg quadruped with 12 degrees of freedom, a list of possible alternatives was tabulated and compared. These motors were sourced under the assumption that their behaviour would closely resemble that mentioned in their datasheet (see [Appendix 7: Tarot Motor Datasheet](#)); the following motor related discoveries were made down the road:

- These are drone motors, peak currents stated along with type of propellers assume active cooling from the convection flow
- These motors during testing overheated at ~15A of continuous current draw with no cooling
- The 5008 version sourced by the team had a very bulky rotor; other alternatives were discovered in the same motor frame weighing ~80g instead of 170g
- The rotor set screws regularly stripped and would cause shaft misalignment / torque transmission issues after replacing shaft

See [Appendix 2: Motor Analysis and Alternatives](#) for an insight into the factors that led to motor choice. The motors were disassembled and their shafts replaced with custom shafts for the actuator. The beginning of the second reduction stage also utilised a custom shaft + pulleys. Shafts were initially Stainless Steel ground to size. From an accessibility perspective, a flat feature for set screws and a shoulder for retaining rings was ground / cut using a rotary tool and the pulleys attached to the shaft using a high strength 2-part epoxy. This setup sufficed for no-load testing but the small pulley on the center shaft failed under load.

A lower reduction ratio that allowed a small pulley with a larger diameter was also tested in order to make the do-it-at-home shafts work; ~7.5:1. But the small pulley still failed and there was a torque tradeoff with already weak motors. A team decision was made to source the shafts + pulleys from a 3rd party manufacturer. (+\$)

Initial Shaft minus Pulley



Final Motor Shaft



Figure 9. Proposed vs Final Shaft

The figure below outlines problems that popped up with the actuator system apart from the isolated motor issues mentioned above:

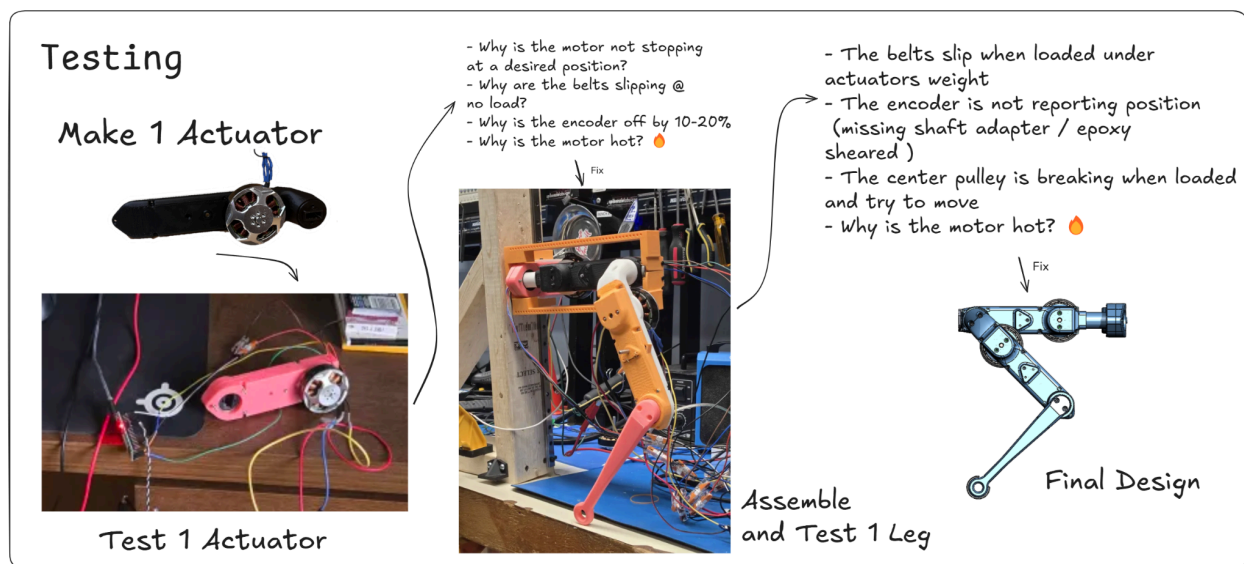


Figure 10. Issues faced during Actuator Testing

The assumed peak current for the motors was never observed in reality and this meant driving heavier motors at lower torque output. This coupled with the unreliability of the motors after multiple disassembly / assembly cycles meant the quadruped needed to lose some weight. There were two proposed paths to achieve this:

- Lose the 4 adduction degrees of freedom
- The new driver stack is offloaded from chassis

See [Appendix 6: Weight Analysis post ODrive](#) for weight loss associated with each option. During this phase the team also entertained the idea of sourcing quality motors from

ODrive; but given this was 6 months into the project and the motors were the backbone of every design decision the following reasons deterred the team from going that route:

- The ODrive motor was rated for 30A (no cooling) but was ~110g heavier than our current motors. Just like the motor alternatives at the start of the capstone; a big chunk of the torque increase is eaten up by the increase in the mass of the motor
- The ODrive motor had a 5mm shaft and all our bearings were for 6mm / 4mm shafts
- Reworking actuator shell for the new motor and the center plates to accommodate the adduction motors

The cost of change at this point was very high which is a given how pivotal the motors are to the entire design process and swapping them out creates many changes downstream as well. A lesson learnt in buying once and crying once. Even with the offloaded electronics; some motors started returning phase issues and worked only half the time. Due to time constraints and a lack of spare motors; the team decided to give up on the 4 adduction motors since this meant:

- Couple of spare motors for the hips and knee
- Additional weight lost from quadruped

The shells being FDM printed did bring about some tolerancing issues with respect to shaft alignment, pulley distance / tensioner location, issues that were brushed out over many iterations. These would be non-issues if a more accurate manufacturing process like SLS / SLA was used.

Appendix

Appendix 1 : Belt Sizing

Using the MISUMI belt sizing guide

Load Calculations:

300kV at approximately 24V ~ 7200rpm for the small pulley

The tarot motor at the input has a maximum torque of 0.5NM at the input. Assuming this maximum torque at the motors maximum velocity (NOT PHYSICALLY POSSIBLE BUT WORST CASE SCENARIO xD)

Power = TorqueW(rpm)pi/30 = 0.376W

Ko = 1.6 (Assuming an application similar to a milling machine for 3 hours a day)

Kr = 0.3

Ki = 0.1

K total = 2

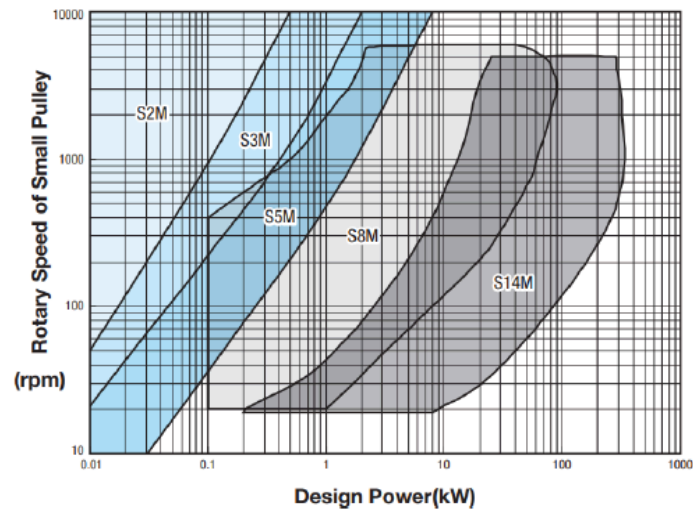
(The K values are just factor of safeties for overload, reduction ratio and tension provided by consumer for choosing a belt)

Design Power: 0.75kW

Which for a worst case scenario falls barely on the edge of the s3m recommended use case.

Hence it is a good idea to try it out in the actuator as a cost saving exercise.

Table 19. Selection Guide Table 2 (S_M series)



From among the commonly available belts on aliexpress like the gt2, trapezoid, htd and s3m belts; the main 2 alternatives for our use case are the htd and s3m varieties. The htd is slightly outside our rpm values while the s3m satisfies both our power and speed needs. (Assuming appropriate tensioning, which will need to be tested for.)

All pulleys in the solo12 design are based on the at3 tooth and have been modified with s3m alternatives for testing. Only information regarding the s3m profile is:

Figure 12. Belt Sizing Notes

Appendix 2: Motor Analysis and Alternatives

(no addition) shared

W = $2.8 \times 10 = 28 \text{ N}$

$T_{\text{req}} = r \cdot F : 0.38 \times 28 = 10.64 \text{ N}\cdot\text{m}$

$T_{\text{req}} = 5.32 \text{ N}\cdot\text{m}$ if only one actuator holds everything

If 14N shared by the 2 actuators loading modelled by

$T_{\text{HFE}} / (0.22) + T_{\text{KFE}} / (0.38) = 14 \text{ N} \Rightarrow$ can be modelled using limits $T's \leq 2.4 \text{ Nm}$ which is 50/12 T's.

⊗ 11th ground = max loading scenario as point of application load farthest from COG $\therefore$ large lever arm

we see that for a static analysis a 2.8 kg quadruped should be able to technically lift itself using 4 of its actuators w/ a max torque of approx. 2.3 Nm

addition slightly complicates hand calcs as becomes 3D problem

SRDT

also bear of proximity to 18th even w/ 2 feet grounded req. $14 \times 0.1 = 1.4 \text{ Nm}$ which is lightwork

$x = 10 \text{ cm}$

$y = \sqrt{10^2 + 22^2} = 24.17 \text{ cm}$

$z = \sqrt{(10 + 16 \cos 45)^2 + (22 + 16 \sin 45)^2} = \sqrt{491.4 + 1049.2} = 39.3 \text{ cm}$

$\therefore \frac{T_{\text{qu}}}{0.1} + \frac{T_{\text{Hn}}}{0.242} + \frac{T_{\text{K}}}{0.393} = 14 \quad T's \leq 2.4$

T_{an} can be ignored as does not physically lift quadruped + move weightage on other 2 dof.

The solution space is much tighter than 8 dof loading but still possible:

WRT worst case static scenarios for a 2.8 kg 12 dof quadruped aiming for atleast a 2.4 Nm actuators will ensure the dog can make attempt to locomote

Figure 14. Motor Sizing Notes

Solo12 testing logs maximum torques for a walking gait

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Joint 0: Max Torque = 0.3373 Nm
Joint 1: Max Torque = 1.1235 Nm
Joint 2: Max Torque = 2.1267 Nm
Joint 3: Max Torque = 1.3014 Nm
Joint 4: Max Torque = 0.7860 Nm
Joint 5: Max Torque = 1.7960 Nm
Joint 6: Max Torque = 0.2281 Nm
Joint 7: Max Torque = 0.7339 Nm
Joint 8: Max Torque = 0.3751 Nm
Joint 9: Max Torque = 1.3256 Nm
Joint 10: Max Torque = 0.9044 Nm
Joint 11: Max Torque = 0.3786 Nm
Press any key to continue . . . |

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Figure 15. Solo12 Testing Logs Max Torques

Proposed alternatives

	Operating Voltage (v)		Kv		Max Rated Current (A)		Stall Torque (Nm)		Ease of Shaft Removal		Weight (g)		Price (CAD\$)		Scores
T-motor MN4004 (Baseline)	(24)		(300)		(12.3)		(0.26)		(7)		(54)		102		
SunnySky V3506	24	10	400	7	12.5	7	0.26	7	6		85	7	49	6	51
Tarot 5008	24	10	340	9	18.2	10	0.44	9	7		178	4	29	8	57
Mitoot 5010	24	10	360	8	12.5	7	0.29	7	8		80	7	19	10	57

Figure 16. Motor Alternatives Assessment

Appendix 3: Center Plate Analysis

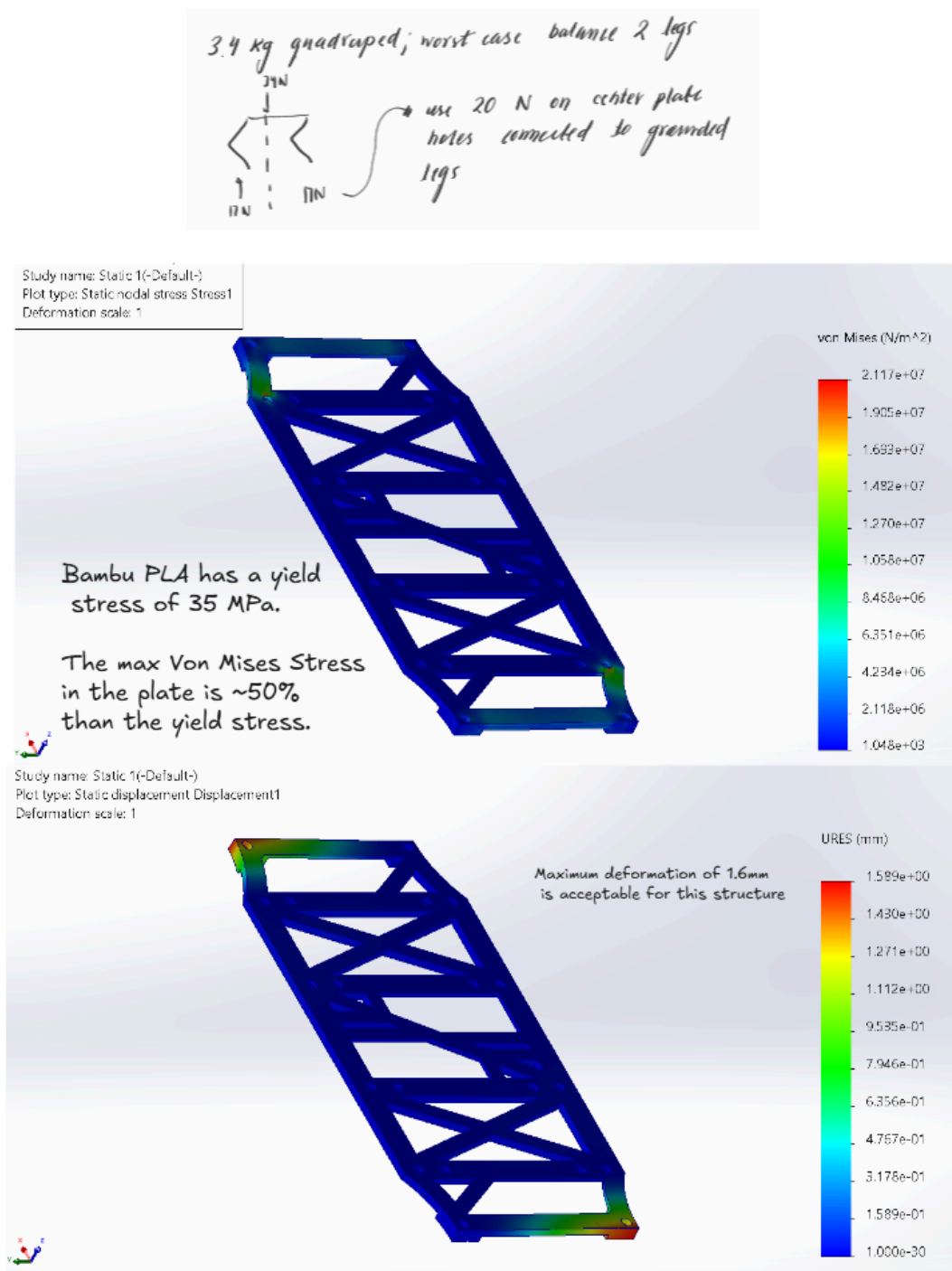


Figure 17. Center Plate Stress and Displacement Contours

Appendix 4: Small Pulley Analysis (Center Shaft)

Center pulley $15A \approx 0.31 \text{ Nm @ Motor} \therefore 0.93 \text{ Nm at } 2^{\text{nd}} \text{ stage}$
 $\approx 5 \text{ mm radius} \therefore F = \frac{0.930}{0.005} = \underline{186 \text{ N}}$

Figure 18. 10-tooth Center Pulley Bending Force

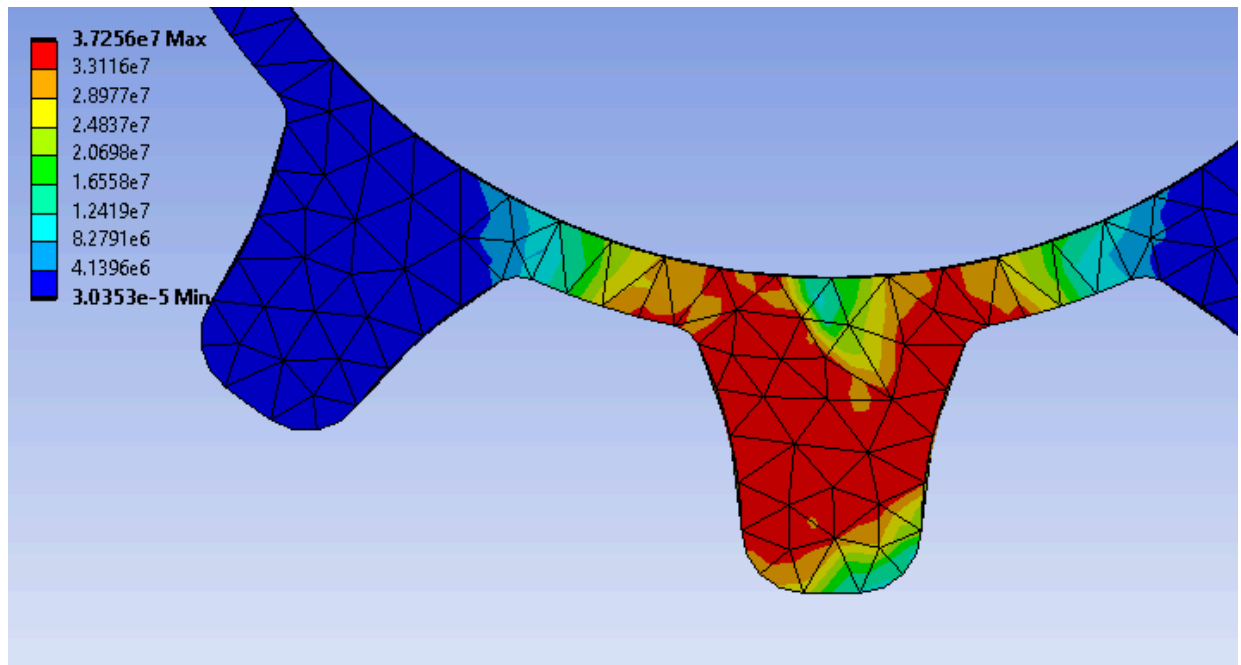


Figure 19. 10-tooth Center Pulley Stress Contour

An ABS-like resin from formlabs has a post-cure yield stress of 46 MPa. The maximum *von Mises* stress at the base of the teeth is 23% less than the yield stress and hence it was considered as a viable accessible option to getting the pulleys manufactured.

Appendix 5: Newton CAD Breakdown

Chassis Assembly

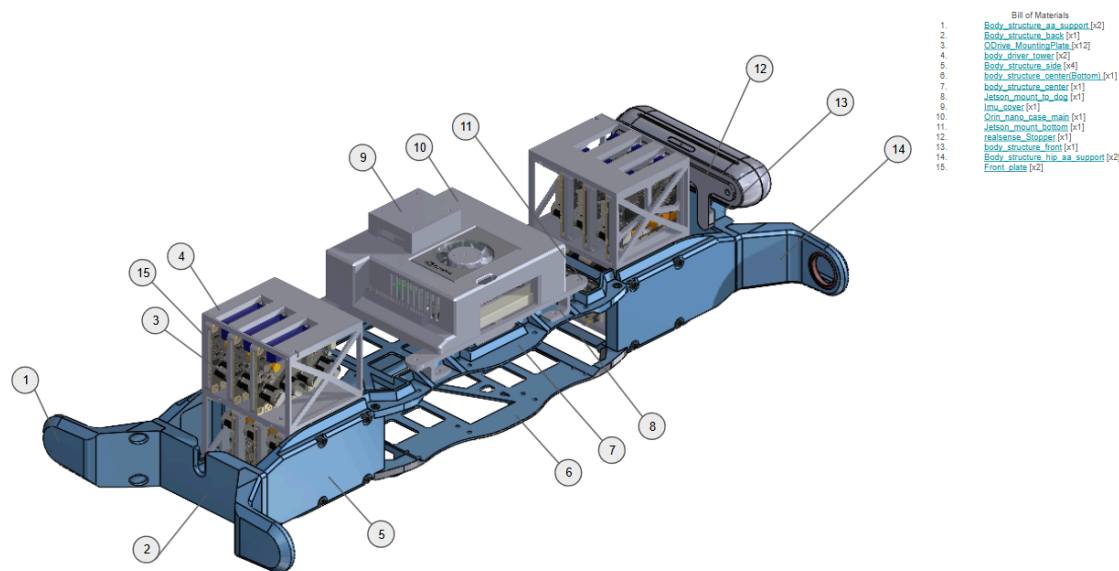


Figure 20. Chassis Assembly

Leg Assembly

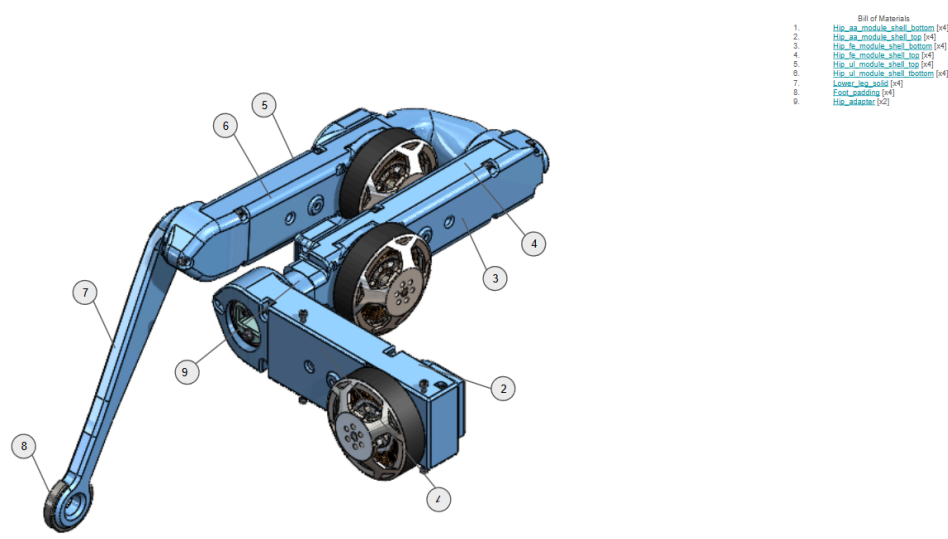


Figure 21. Leg Assembly

Actuator

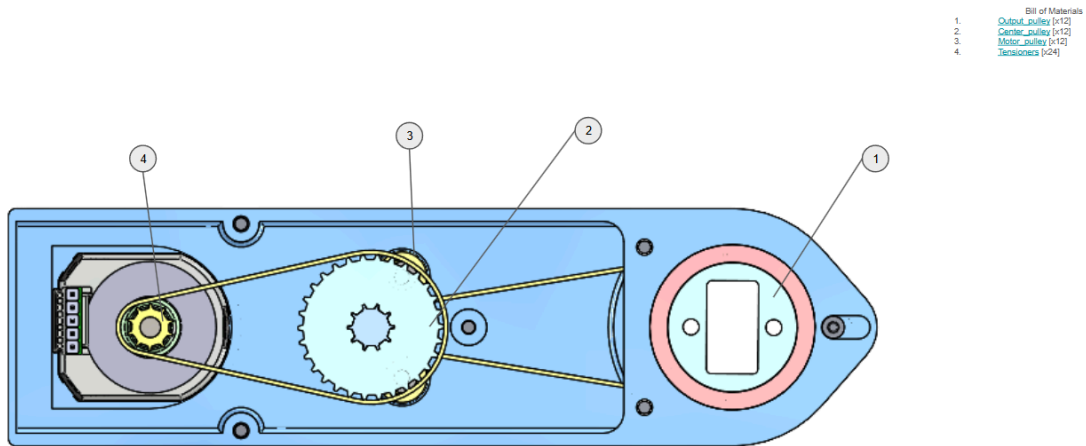


Figure 22. Actuator Assembly

Simplified collision meshes with coordinate systems for joints

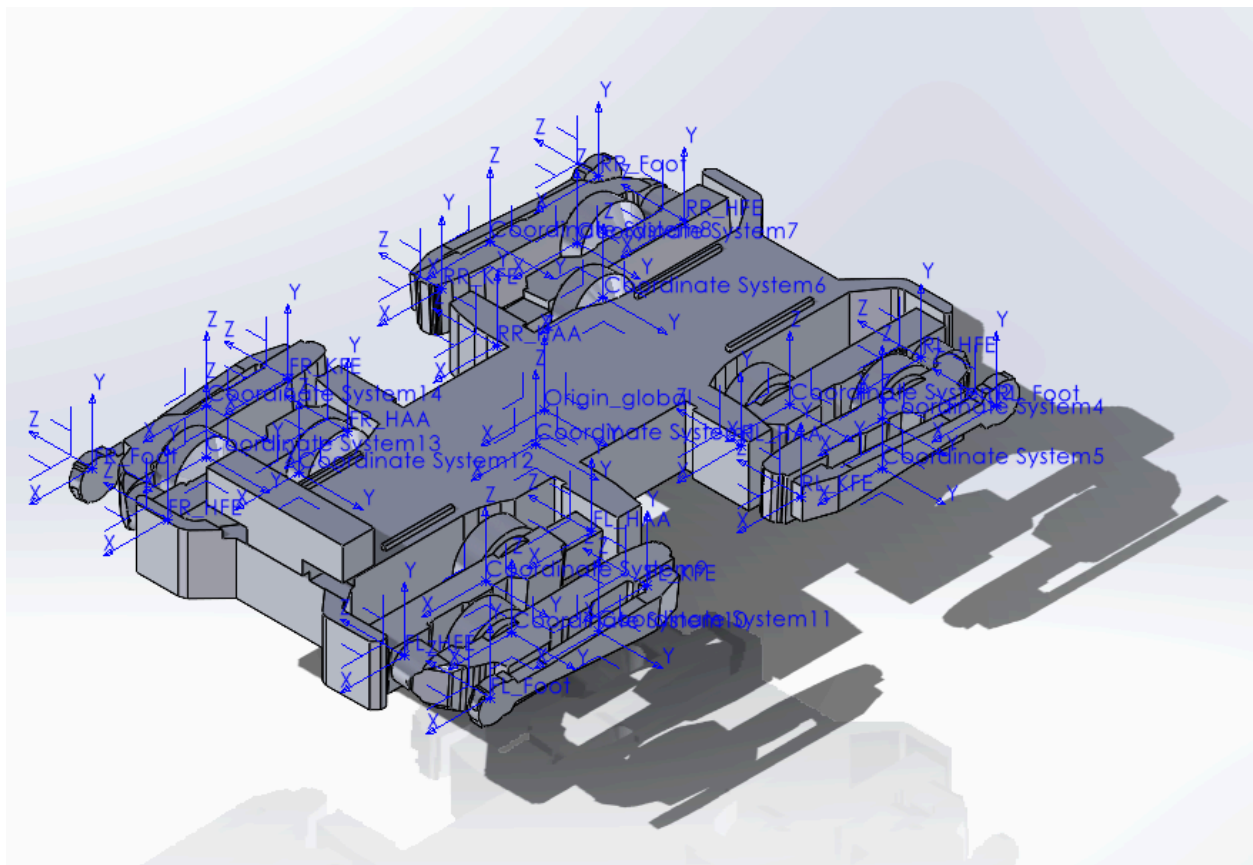


Figure 23. Collision Meshes + Coordinate System for Joints

Appendix 6: Weight Analysis post ODrive

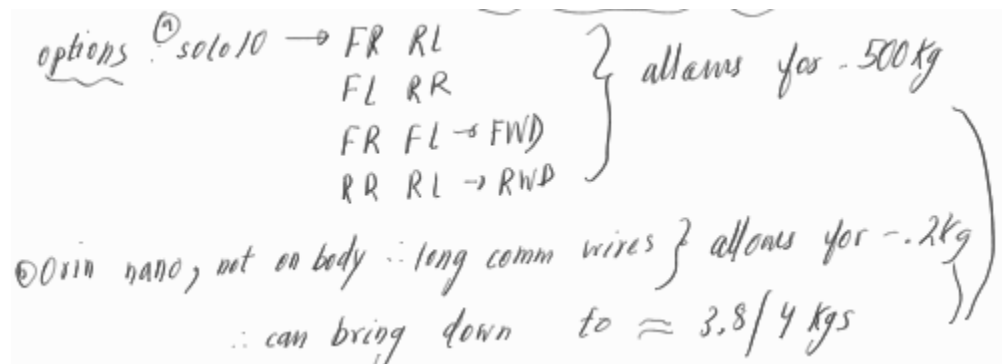
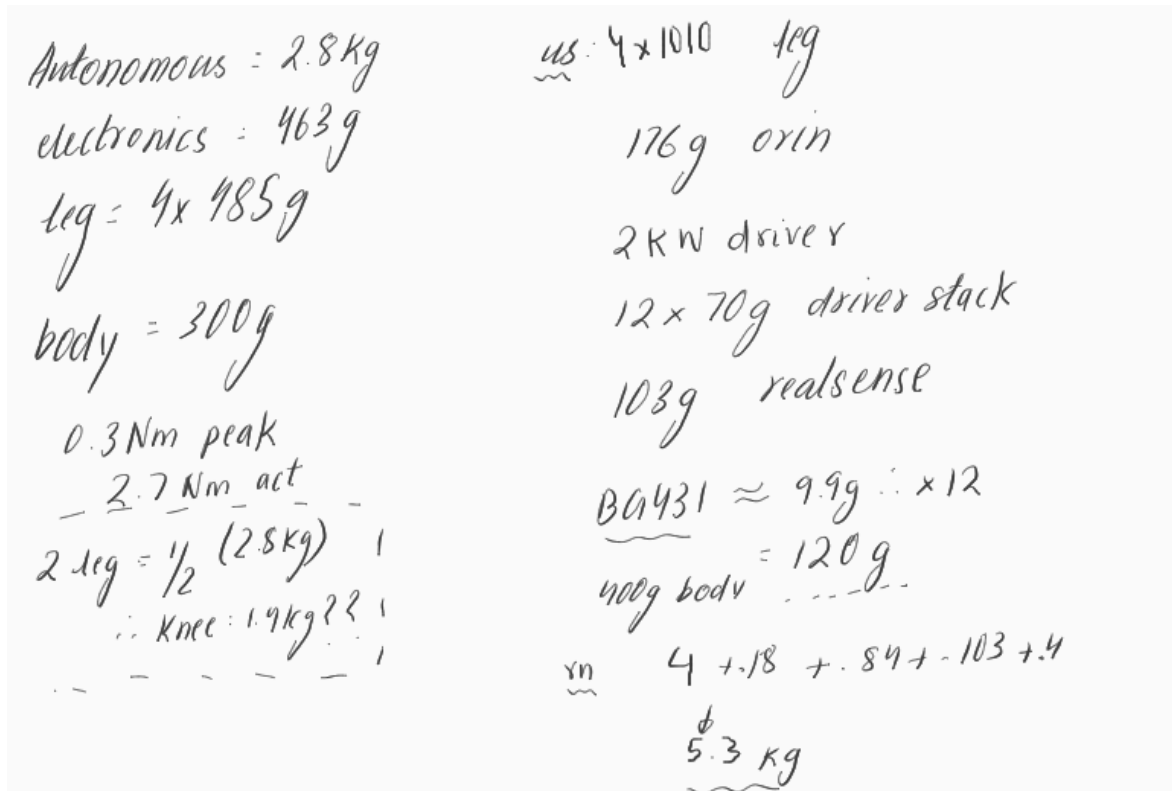


Figure 24. Post ODrive Weight Analysis + DoF Decision

Move to 10 degrees of freedom, frees up $\sim$ 0.5kg and drops weight down to 4.8 kg. Each actuator removed frees up 250g.

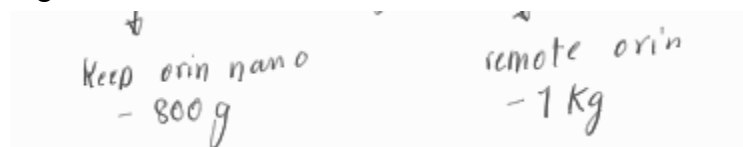
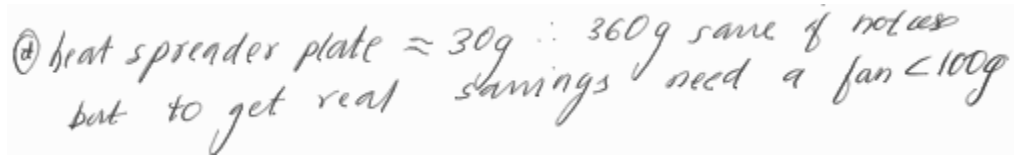


Figure 25. Jetson Decision

So with 8 degrees of freedom the quadruped would weigh 4.3 kg.

Offloading the Jetson from the dog quadruped frees up ~180 grams and brings the weight down to 4.1 kg.



④ heat spreader plate $\approx 30g \therefore 360g$ same if not used
but to get real savings need a fan $< 100g$

Figure 26. ODrive Decision

Just removing the heat spreader plates from the ODrives in the driver stack frees up ~360 grams
But then they require active cooling dropping the weight down to 3.8 kg.

But if offloading the entire driver stack (840 grams) instead this would bring the final quadruped weight down to 3.3kg.

Appendix 7: Tarot Motor Datasheet

Motor	Propeller	6S	22.2V	Throttle(%)	0	17	34	51	68	85	100
Tator 5008 kv340	Falcon 1855CF			AMP(A)	0	0.7	3.5	10.8	18.7	21	21.3
				Thrust(KG)	0	0.282	0.930	1.990	2.800	3.023	3.066
				Efficiency(G/W)	0.00	18.15	11.97	8.30	6.74	6.48	6.48

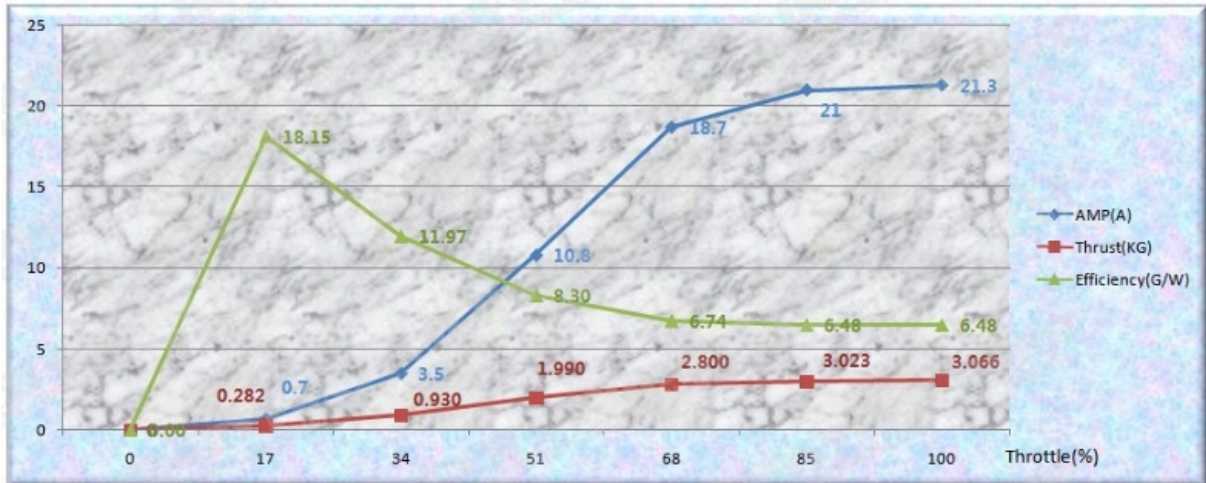


Figure 27: Tarot 5008 340kv Datasheet

Appendix 8: Jigs

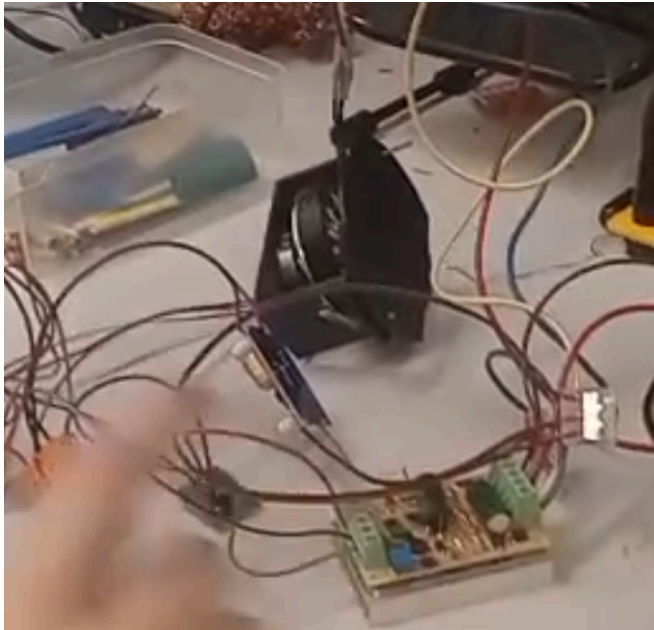
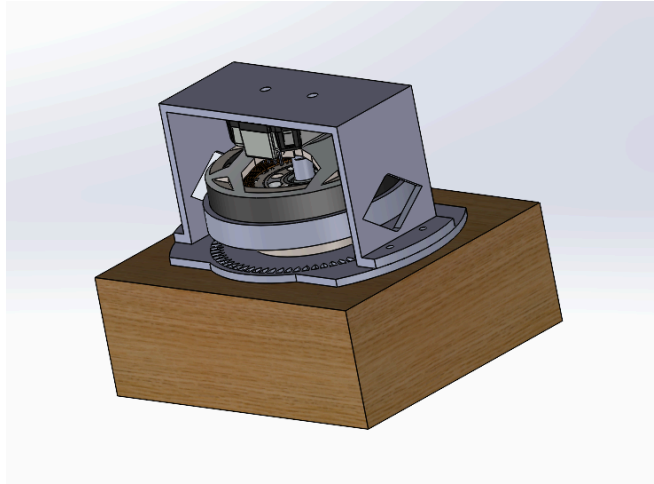


Figure 28: ESP32 Control Loop Testing

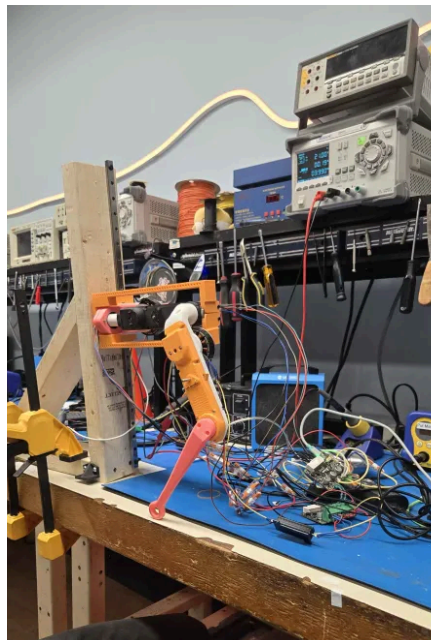


Figure 29: Leg Testing Jig

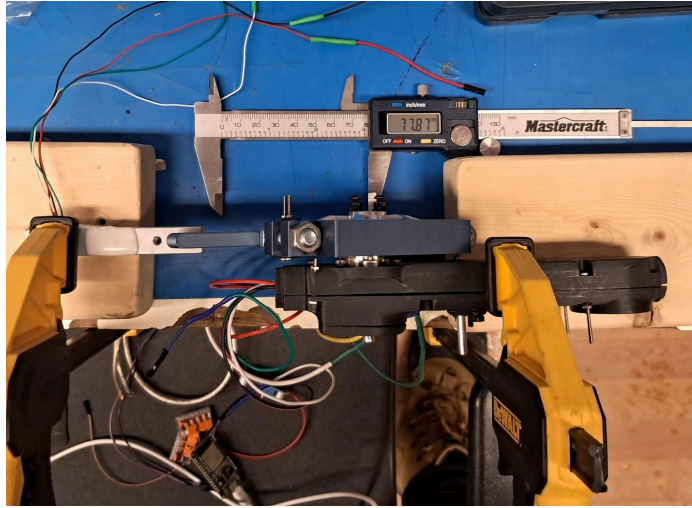


Figure 30: Prony Brake with Load Cell

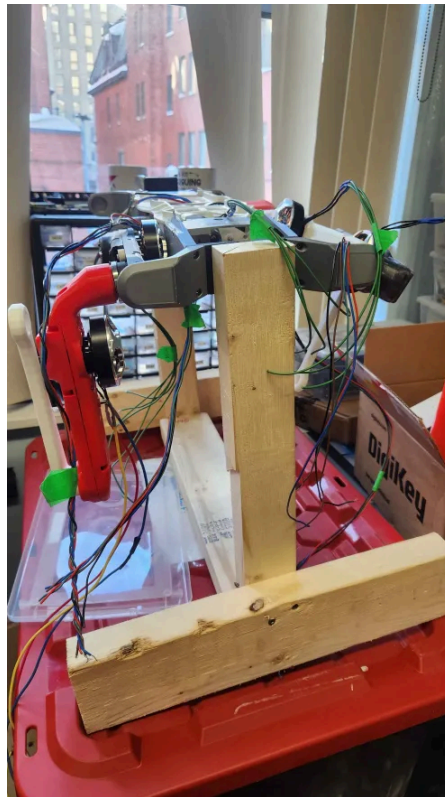


Figure 31: Storage + Calibration Jig